

Lakshmi Srikanth Polavarapu

Bangalore, India

AI Engineer | RAG & Agentic AI | LLMOps

Individual Contributor | Self-Learner

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PROFESSIONAL SUMMARY

AI Engineer with 1.5+ years of hands-on experience building and shipping production-grade GenAI systems at Cisco. Specialized in end-to-end RAG pipelines, Agentic AI with MCP tool orchestration, vector database engineering (ChromaDB, Milvus), and on-premise LLM deployment on GPU clusters. Proven track record of taking AI systems from R&D prototype to production products adopted by engineers across Bangalore, San Jose, and CRDC. Currently exploring on-premise deployment and domain adaptation of **Gemma 31B** on Cisco GPU infrastructure using domain-specific platform logs.

EDUCATION

•Bachelor of Technology in Electrical and Electronics Engineering

Raghu Engineering College, Visakhapatnam

2020 – 2024

CGPA: 7.5

EXPERIENCE

•AI Engineer – Contingent (via Adecco)

Dec 2025 – Present

Cisco

Bangalore

- Building **AI Agent systems** for engineering workflows, focusing on agent orchestration, tool use, multi-step reasoning, and autonomous task execution across internal Cisco use cases.
- Developing and experimenting with **coding agents** using **RLM (Recursive Language Model) concepts**, exploring recursive reasoning, context management, sub-agent execution, and iterative task decomposition for complex software engineering workflows.
- Working on **LLM inference engineering** in depth, including on-premise model serving, inference optimization, GPU utilization, serving architectures, and evaluation of frameworks such as **vLLM, SGLang, and LiteLLM**.
- Co-built a custom **MCP server from scratch** for internal Cisco hardware schematic and layout analysis, wrapping **Codex** as the LLM backbone to parse, reason, and answer queries over design files.
- Leading R&D exploration of **Gemma 31B** for on-premise deployment on Cisco GPU servers, curating and preparing domain-specific datasets from platform logs for model adaptation and evaluation.

•Software Engineer Apprentice

Nov 2024 – Nov 2025

Cisco

Bangalore

- Architected and shipped a **production-grade RAG system** from scratch — starting with basic LLM chatbots, evolving through a ChromaDB-based prototype, and ultimately deploying on **GPU-accelerated Milvus** on Cisco's internal servers, containerized via **Docker and Kubernetes**.
- Product adopted enterprise-wide across **Bangalore, San Jose (SJC), and CRDC** — assisting 40+ diagnostic engineers in debugging complex platform logs with **90%+ semantic retrieval accuracy**.
- Presented the system to **VP-level stakeholders** at Cisco internal hackathons and technology roadshows (Sep 2025); project recognized by the R&D team and transitioned into a formally supported enterprise product.
- Engineered the full RAG pipeline: multi-format document ingestion, intelligent chunking, **HuggingFace embedding generation**, and vector indexing with **Elasticsearch** and **Milvus** for natural-language log debugging.
- Integrated **LangChain** and **Azure OpenAI** for semantic understanding, context-aware retrieval, and AI-driven root cause analysis across multi-platform logs (1K–300K+ lines per file).
- Built a **FastAPI-based dashboard** for the DIAGS automation team; contributed to automating nightly regression scheduling, command file updates, and automated test-result email pipelines.

PROJECTS

•Enterprise RAG System — Prototype to Production (RILDS)

LangChain | Milvus | ChromaDB | HuggingFace | Azure OpenAI | Docker | Kubernetes | GPU Servers

- Ran controlled benchmarking experiments comparing FAISS, Milvus, and ChromaDB across latency, precision, memory footprint, and retrieval correctness on logs ranging from 1K to 300K+ lines.
- Selected ChromaDB for rapid prototyping (best stability/dev-speed); migrated production to **Milvus with GPU acceleration** for high-throughput, scalable vector search at enterprise load.
- Containerized the full stack with **Docker and Kubernetes**; deployed on Cisco's internal GPU cluster for enterprise-grade availability and horizontal scalability.
- **Outcome:** 90%+ semantic retrieval accuracy; product now live and actively used across BLR, SJC, and CRDC engineering teams.

•AI Agent System with MCP Tool Orchestration

MCP (Model Context Protocol) | OpenWebUI | Codex | Function Calling | Schematics Analysis

- Contributed to an internal Cisco agentic AI workflow leveraging **MCP** for structured tool use, enabling the agent to autonomously call engineering functions, retrieve data, and execute multi-step workflows.

- Integrated **OpenWebUI** with the existing CLI agent, creating a production-ready browser interface that dramatically lowered the barrier to adoption for non-terminal users.
- Co-designed and built a **custom MCP server for internal Cisco hardware schematic and layout analysis** — using Codex as the LLM wrapper to parse design files and answer engineering queries with contextual reasoning.
- AI Fine-Tuning Agent — NVIDIA NeMo Infrastructure (Ongoing)**
NVIDIA NeMo | Agentic AI | Fine-Tuning | Domain Adaptation | GPU Inference | Cisco Internal LLM
 - Building an autonomous **fine-tuning agent** that uses the **NVIDIA NeMo** technology stack to automate the end-to-end fine-tuning workflow for Cisco’s internal LLMs on domain-specific data, currently focused on platform logs.
 - Developing the agent to automate dataset preparation, fine-tuning configuration, training execution, evaluation, and iterative model experimentation using **NVIDIA NeMo** infrastructure and Cisco’s internal GPU environment.
- ShipAI — Offline-First AI Engineering Platform [GitHub]**
FastAPI | LangChain | ChromaDB | BM25 | LangGraph | MCP | Ollama | Redis | Celery | Docker | PyWebView
 - Built a full-stack AI platform that generates production-ready AI project codebases (RAG, Multi-Agent, Data Analyzer, LLM Fine-Tuner) from a single CLI command — designed to solve real engineering gaps observed while building and maintaining enterprise AI at Cisco.
 - Implemented advanced “**RAG for Builders**” patterns beyond basic retrieval: **query reformulation**, **PII masking**, **hybrid search** (ChromaDB vector + BM25 sparse), pre-configured with **7 enterprise infra patterns** (Rate Limiting, Celery workers, Prometheus/OpenTelemetry, Locust, Circuit Breakers, API Gateway, Auto-Scaling scaffolds).
 - Shipped “**Agentic AI for Builders**” template with production-grade agent infra: **LangGraph state persistence**, trajectory tracing, infinite-loop protection, **human-in-the-loop** approval endpoints, and native **MCP integration**.
 - Deployed via **3 interfaces**: CLI (engineers), REST API/SDK (enterprise), and a fully standalone **1-click Desktop App** packaged with PyWebView + PyInstaller — no terminal required for non-technical users.
 - Fully offline-first via **Ollama local LLMs**; also supports cloud APIs (OpenAI/Anthropic) for scale. Containerized with **Docker Compose**; open-sourced on GitHub.
- AI Prompt Generator for Automated Log Analysis**
Cisco GPT API | Python | Regex Parsing | Dynamic Prompt Engineering
 - Built a Python tool that auto-analyzes raw log files and dynamically generates optimized prompts — enabling engineers to get high-quality AI responses without manual prompt crafting.

TECHNICAL SKILLS

Programming Languages: Python, C, Java

GenAI & LLM Systems: RAG, Agentic AI, LLM Fine-Tuning, Prompt Engineering, MCP, Function Calling, OpenWebUI, Transformers, Hybrid Search (BM25 + Vector), Query Reformulation

Vector Databases: Milvus (Production), ChromaDB, FAISS, Elasticsearch

Frameworks & Libraries: LangChain, LangGraph, HuggingFace, FastAPI, TensorFlow, PyTorch, scikit-learn, Flask, Streamlit

LLMOps & Infra: Docker, Kubernetes, Ollama, Redis, Celery, Prometheus, OpenTelemetry, vLLM, SGLang, LiteLLM, GPU Server Deployment, Azure OpenAI, On-Premise LLM Serving

Databases: MongoDB, MySQL

Dev Tools: Git/GitHub, VS Code, PyCharm, Jupyter, REST APIs, HTML/CSS/JS

CS Fundamentals: Data Structures, Algorithms, OOP, Networking, Software Design

CERTIFICATIONS

- IBM AI Engineering Professional** (Coursera)
- CCNA Certified** (Cisco Certified Networking Associate)
- LangChain – Develop LLM Applications** (Udemy)

ACHIEVEMENTS

- RAG System Adopted Enterprise-Wide** — R&D product now live across BLR, SJC & CRDC engineering orgs at Cisco
- SVP-Level Presentation** — Presented RAG system architecture and roadmap at Cisco internal technology roadshow (Sep 2023)
- Hackathon Finalist** — Team DIAGS, RAG-based AI Assistant (Cisco Bridge IT Hackathon)
- 1st Prize** — Poster Presentation on GSM-Based Energy Meter (Faraday’s Memorial Event, 2023)
- Research Publication** — EV Scheduling & Power Loss Minimization, Journal of Computing, 2023